Week 5

MICROECONOMIC THEORY
ECON 323 502/503

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Review

To study how a price change effects quantity demanded of a good, we have studied the concepts of substitution effects and income effects.

1. The response to a change in the relative prices of goods is called the **substitution effect**.

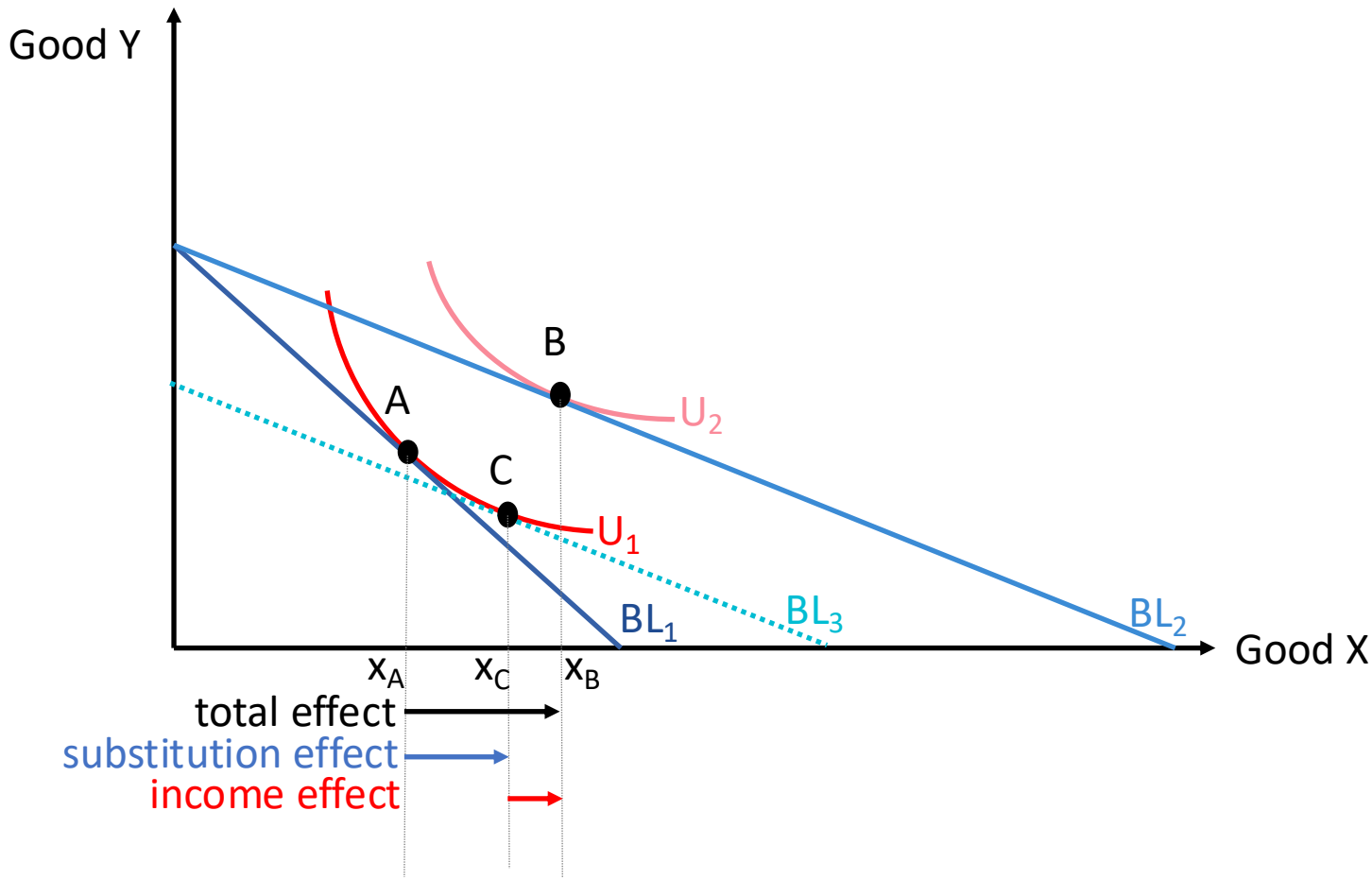
2. The change in consumption resulting from the change in real purchasing power is called the **income effect**.

Review

We have learned how to visualize these effects as well as the total effect in a graph for a normal good, inferior good with small income effect, and Giffen good (when price goes down, the quantity demanded decreases).

1. Plot auxiliary budget line (BL): parallel to ultimate BL, tangent to the initial indifference curve.
2. Identify an auxiliary utility-maximizing market basket.
3. Substitution effect: initial demand → auxiliary demand
4. Income effect: auxiliary demand → ultimate demand

Normal Good Example -- P_X decreases



Market Demand

Market demand curve: Curve relating the quantity of a good that <u>all consumers in a market will buy</u> to its price.

Example: Suppose there are 10 male buyers and 10 female buyers in the market,

- Each male buyer has a demand curve: $Q_{male}(P) = 3/P$.
- Each female buyer has a demand curve: $Q_{female}(P) = 2-2P$.

What is the function form of the market demand curve?

Solution: $Q_m(P) = 10*(3/P) +10* (2 - 2P)= 30/P + 20 - 20P$.

Elasticity of Demand

The price elasticity of demand describes the percentage change in quantity demanded of one good caused by every percentage change in price of the good.

Mathematically, it is defined by $E_P = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}}$,

- $\frac{\Delta Q}{Q}$ is the percentage change in quantity demanded
- $\frac{\Delta P}{P}$ is the percentage change in price

Example: Suppose when the price increases by 5%, the consumption of burger decreases by 10%, the price elasticity of burger is equal to $\frac{-10\%}{5\%} = -2$.

Elasticity of Demand

- When E_P is less than 1 in absolute value, demand is **inelastic**. E.g., $E_P = -0.5$.
- Consumers' total expenditure on the product/firm's revenue increases when the price increases.

- When E_P is greater than 1 in absolute value, demand is **elastic**. E.g., $E_P = -2$.
- Consumers' expenditure on the product /firm's revenue decreases when the price increases.

- When $E_P = -1$, demand has **unit** elasticity.
- Consumers' total expenditure on the product /firm's revenue does not change when the price increases.

Elasticity of Demand

How to compute price elasticity of (individual or market) demand analytically?

$$E_P = \frac{\frac{\Delta Q}{Q}}{\frac{\Delta P}{P}} = \frac{\Delta Q}{Q}\frac{P}{\Delta P} = \frac{\Delta Q}{\Delta P}\frac{P}{Q} = Q'(P)\frac{P}{Q}.$$

Elasticity of Demand

Example: When the market demand curve is given by $Q(P) = \frac{100}{P}$ and price $P = 2$, what is the price elasticity of the market demand.

Solution:
Recall $P = 2$.
$Q'(P) = -\frac{100}{P^2}$ and thus $Q'(2) = -\frac{100}{2^2} = -25$.
$Q(P) = \frac{100}{P}$ and thus $Q(2) = \frac{100}{2} = 50$.

$E_P = Q'(P)\frac{P}{Q} = (-25)\frac{2}{50} = -1$.

Elasticity of Demand

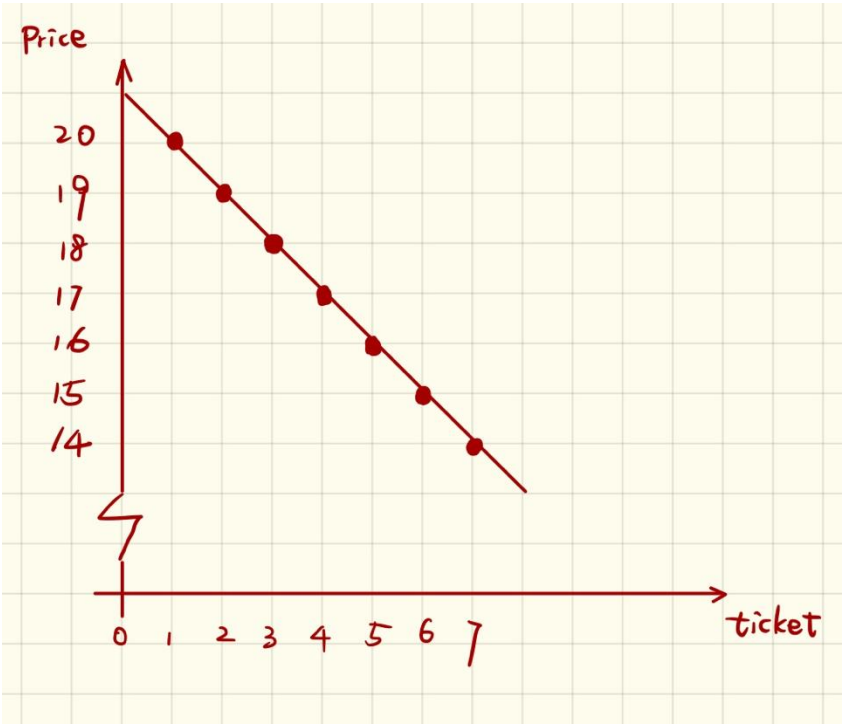
Example: When the individual demand curve is given by $Q(P) = 10 - P$, and price $P = 1$, what is the price elasticity of the individual demand?

Solution:
Recall $P = 1$.
$Q'(P) = -1$.
$Q(P) = 10 - P$ and thus $Q(1) = 9$.

$E_P = Q'(P)\frac{P}{Q} = (-1)\frac{1}{9} = -\frac{1}{9}$.

Consumer Surplus

Demand curves shows the willingness to pay for 1 extra Good X.



Willingness to pay	
1st ticket	$20
2nd ticket	$19
3rd ticket	$18
4th ticket	$17
5th ticket	$16
6th ticket	$15
7th ticket	$14

Consumer Surplus

Surplus of a particular unit of good is the difference between what a consumer is willing to pay for this unit and the price.

1st ticket	\$20
2nd ticket	\$19
3rd ticket	\$18
4th ticket	\$17
5th ticket	\$16
6th ticket	\$15
7th ticket	\$14

Suppose market price is \$14. What is the surplus of each ticket?

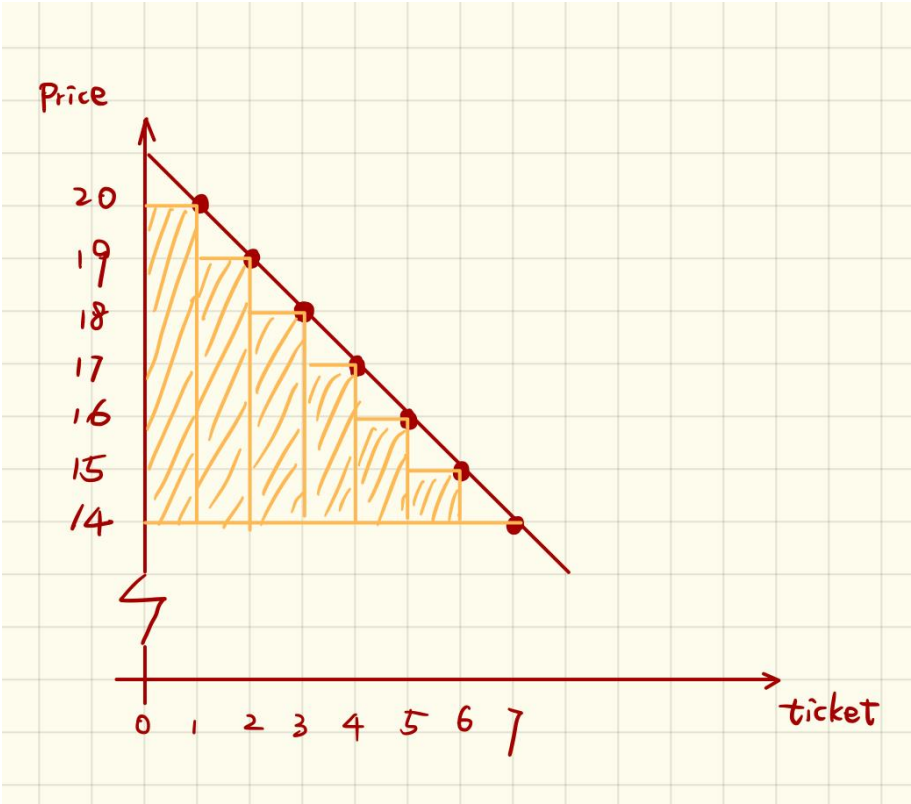
\$6=(\$20-\$14), ← 1st
\$5=(\$19-\$14), ← 2nd
\$4=(\$18-\$14), ← 3rd
\$3=(\$17-\$14), ← 4th
\$2=(\$16-\$14), ← 5th
\$1=(\$15-\$14), ← 6th
\$0=(\$14-\$14), ← 7th

Consumer Surplus

Consumer surplus is the total benefit from the consumption of a product, less the total cost of purchasing it.

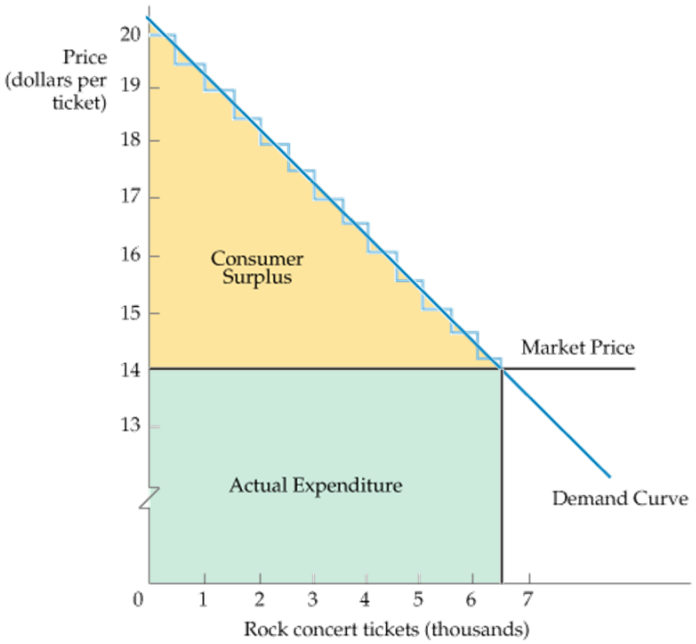
The consumer surplus associated at $14 per ticket is given by the yellow-shaded area:

$6 + $5 + $4 + $3 + $2 + $1 + $0 = $21



Consumer Surplus

Graphically, consumer surplus is measured by the area under the demand curve and above the line representing the purchase price of the good (up to the quantity purchased): the triangle.



Expenditure = purchase price * quantity purchased: the rectangle.

Chapter 3. Consumer Choice.

Preferences:

- Assumptions of completeness, transitivity, more is better
- Utility function, how to read/plot indifference curve/map
- MRS: absolute value of slope of difference curve, economic definition, mathematical definition: $MRS=MU_X/MU_Y$.
- Assumption of diminishing rate of substitution.
- Indifference maps: diminishing MRS, perfect substitutes, perfect complements

Budget constraint:

- Function and meaning of a budget line
- How budget line changes when P_X, P_Y, and/or I changes.

Consumer choice/utility maximization problem:

- Diminishing MRS and Smooth preferences (Default Case): $MRS= P_X/P_Y$ and budget line
- Other cases: use a graph

Chapter 4. Individual and Market Demand

Graphically

- Find price-consumption curve, demand curve, income-consumption curve, Engel curve
- Tell if two goods are complements/substitutes/independent
- Isolate substitution effect and income effect and tell if a good is normal, inferior, or Giffen

Analytically

- Derive the demand curve and Engel curve, tell if some good is normal or inferior

- Derive market demand curve
- Interpret and compute elasticity: elastic, inelastic, unit elastic
- Find willingness to pay/consumer surplus/expenditure